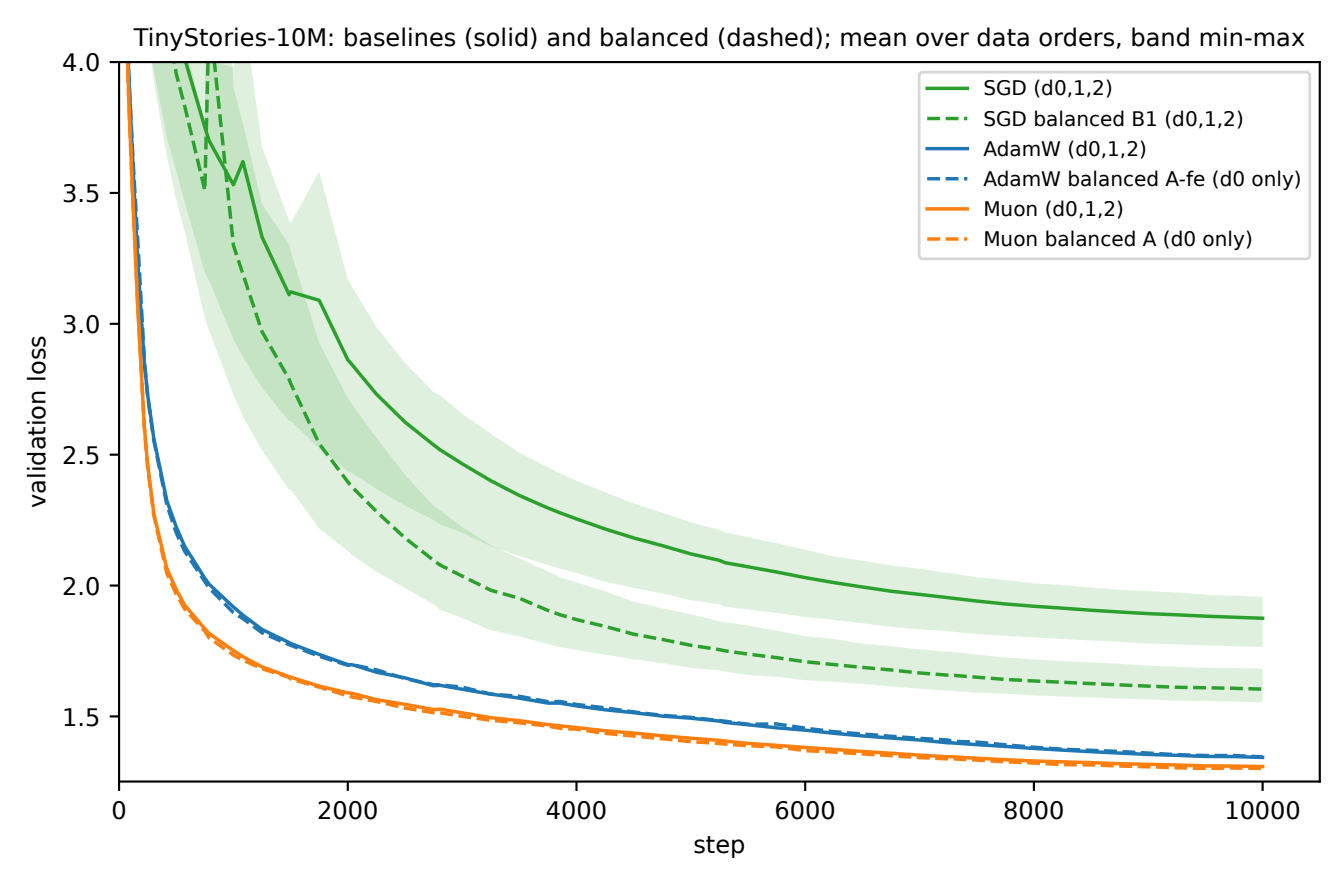
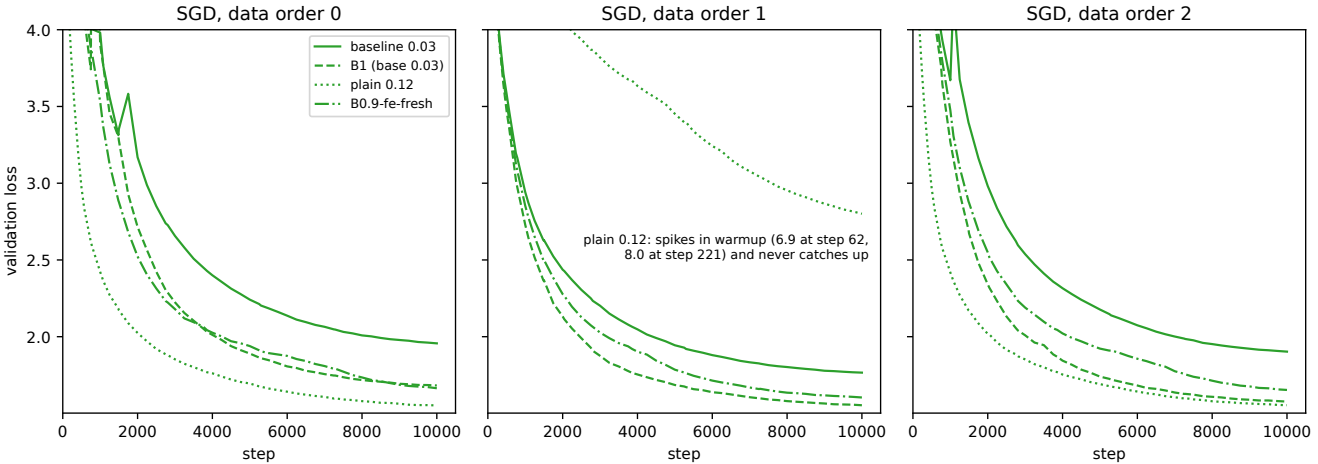
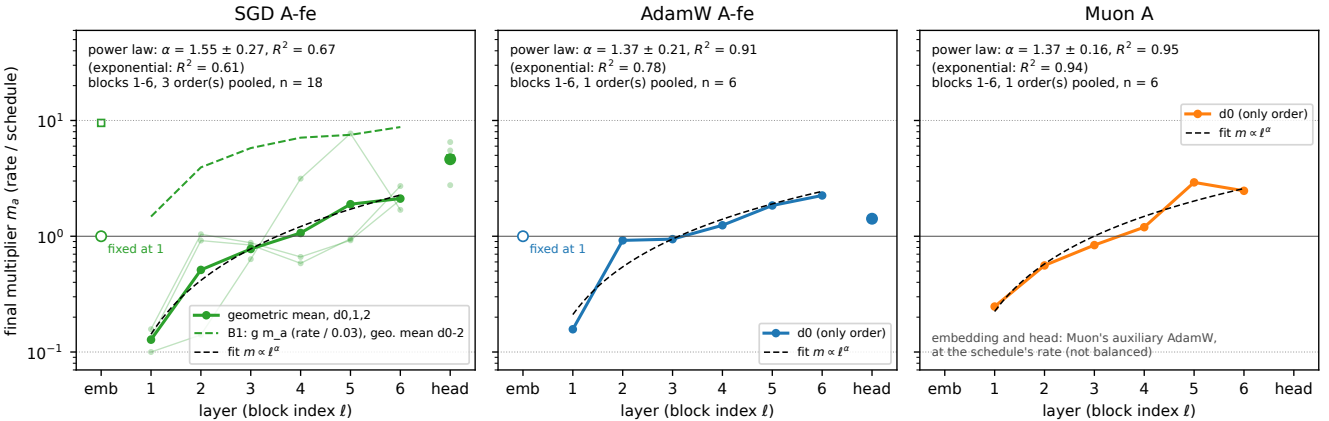
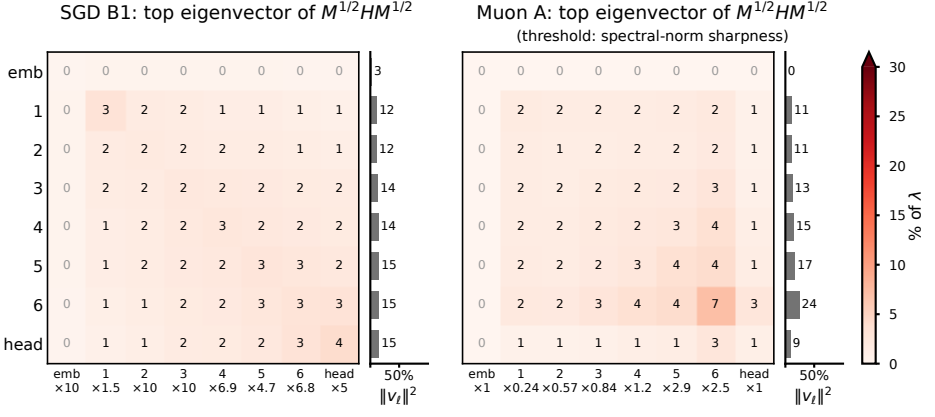

TinyStories-10M: baselines (solid) and balanced (dashed); mean over data orders, band min-max
validation loss
step
4.0
3.5
3.0
2.5
2.0
1.5
0
2000
4000
6000
8000
10000
SGD (d0,1,2)
SGD balanced B1 (d0,1,2)
AdamW (d0,1,2)
AdamW balanced A-fe (d0 only)
Muon (d0,1,2)
Muon balanced A (d0 only)


SGD, data order 0
SGD, data order 1
SGD, data order 2
validation loss
step
baseline 0.03
B1 (base 0.03)
plain 0.12
B0.9-fe-fresh
plain 0.12: spikes in warmup (6.9 at step 62,
8.0 at step 221) and never catches up
4.0
3.5
3.0
2.5
2.0
0
2000
4000
6000
8000
10000



Clear trend in all three: the rate rises with depth, block 1 lowest (0.10-0.25x) and blocks 5-6 highest (about 2-3x); a power law fits better than an exponential. SGD's single orders scatter widely around it (d1: blocks 1-2 at 0.1-0.14x, block 5 at 7.7x); AdamW and Muon have one order each.

Balanced runs, mean over the checkpoints at steps 5000-10000. $M = \mathrm{diag}(m_\ell)$, with m_ℓ each layer's rate multiplier logged by the balancer (under the labels).
Row sums $\|v_\ell\|^2$ by the bars. Left out, no checkpoint with optimizer state (for P): ts10m_balanced/adamw_A-fe_d0.



Balanced runs, mean over the checkpoints at steps 5000-10000. Plain operator (without M); m_ℓ: each layer's rate multiplier logged by the balancer (under the labels).
Row sums $\|v_\ell\|^2$ by the bars. Left out, no checkpoint with optimizer state (for P): ts10m_balanced/adamw_A-fe_d0.

